Aurora Forecasting Test Doc John Ou

Test Microsoft Aurora forecasting abilities in the California central valley and explore fine tuning options. We will look at 3 Aurora models, generate forecasts on the same window as the Healdsburg test, as if we were using these models to inform deployment formation. Then we will compare against real weather data collected during the test and compare with other forecast options.

https://microsoft.github.io/aurora/intro.html

End goal:
We want to be able to generate the best 24hr forecast prioritizing wind speed and direction one day prior to deployment.

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**Test 1: Aurora 0.1° Fine-Tuned**

This is Aurora's highest-resolution checkpoint (~11 km, 1801×3600 global grid), fine-tuned specifically on IFS-HRES analysis. This is the spatial-detail arm of the experiment: it's the only model that resolves fine terrain structure, so it tests whether resolving the Alexander Valley's hills and valley floor actually improves accuracy at your point versus the coarser 0.25° models. Best-in-class Aurora performance when fed HRES analysis, which is the RDA data source here.

*Data:* IFS-HRES analysis, NCAR RDA.

*Step:* 6-hourly.

*Location:* Alexander Valley / Geyserville, 38.6230932, −122.8856340.

*Window:* 7/13–7/16.

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**Test 2: Aurora 0.25° Pretrained**

The un-fine-tuned generalist base model (~28 km, 721×1440), 6-hourly. Useful for two reasons: it's the baseline that shows what fine-tuning and the newer generation actually add, and it's the exact checkpoint you'd fine-tune from, so its off-the-shelf error at your point is the number a local fine-tune would have to beat. Run on the same ERA5 input as Test 3 so any gap between them isolates the value of the 1.5 upgrade alone.

*Data:* ERA5 (Copernicus CDS).

*Step:* 6-hourly.

*Location & window:* same as Test 1.

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**Test 3: Aurora 1.5 (0.25°)**

The newest generation (~28 km, 721×1440), and the only one here with **hourly** output. Beyond time resolution it adds 22 variables the others lack, precipitation, cloud cover, radiation, soil temperature/moisture, 100 m wind, several of which are directly relevant to agriculture. This is the temporal-detail arm: it tests whether hourly steps and the richer variable set verify better at your point than a sharper-but-6-hourly high-res forecast. An ensemble variant also exists if you later want uncertainty spread rather than a single forecast.

*Data:* ERA5 (Copernicus CDS).

*Step:* hourly (6 sub-steps per 6 h main step).

*Location & window:* same as Test 1.

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**GPU Requirements:**

Amazon EC2 G6e Instance - 48 GB GPU

“You will need approximately 40 GB of GPU memory for running the regular model on global 0.25 degree data. Due to some memory optimizations including more use of float16 dtype, Aurora 1.5 should run on a GPU with 32 GB of memory.” from Microsoft usage guidelines